OS TOOLS

<NUKE4ALL>

MIDI Remote Control Software



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Welcome!

Congratulations and thank you for purchasing this novel OSXTools software for your Hartmann Music <Nuke> USB Remote. With the <Nuke4all> application you can use the <Nuke> remote in conjunction with any device that is capable of receiving MIDI control change commands, such as many popular VST and Audio Unit plug ins and instruments. What's more, in combination with your favorite harddisk recording/MIDI sequencing software you can even record automated parameter movements of more than one parameter at one time directly from the <Nuke>. Custom assignable control change numbers let you set up your <Nuke> to fit your setup and working preferences and all settings are remembered across sessions so you are always "ready" when starting up the program.

How does it work?

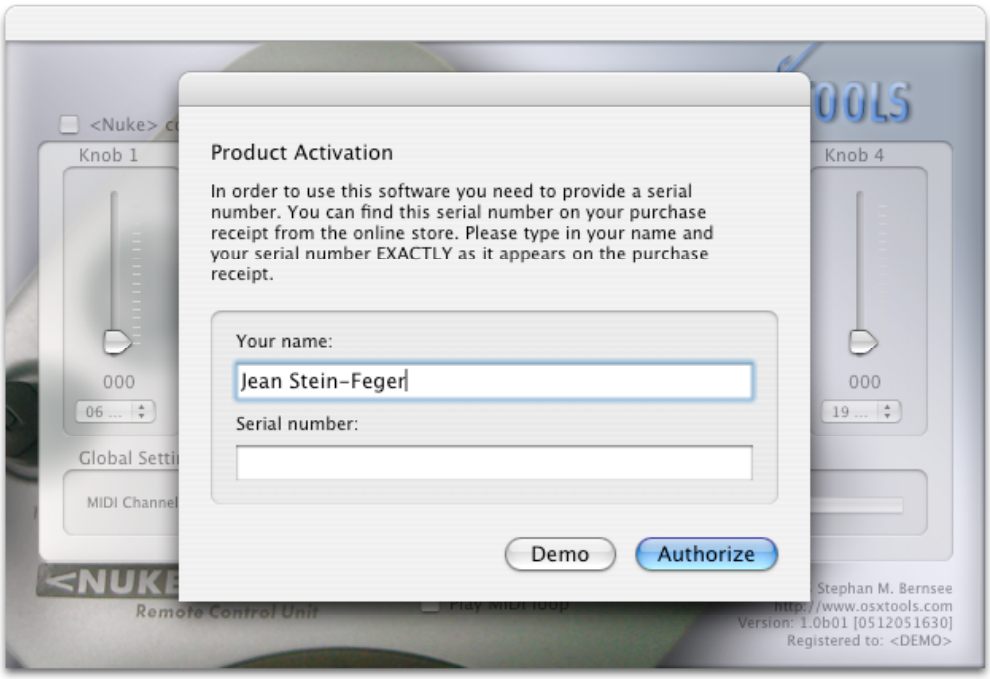
The <Nuke> is an USB device that is attached to any USB port of your Macintosh. Originally developed by Hartmann Music as remote control for their NeuronVS software synthesizer the <Nuke4all> application lets you use this remote for any other synthesizer by translating the knob and stick movements into MIDI messages. For this to work, the <Nuke4all> application needs to be running in the background and your <Nuke> needs to be plugged in to any USB port on your computer.

Installation and Setup

Installation and setup couldn't be easier. No driver installation is required and no low-level system tweaking is necessary to run <Nuke4all>. Thanks to MacOS X, installation and use of the program is as easy as double-clicking the application.

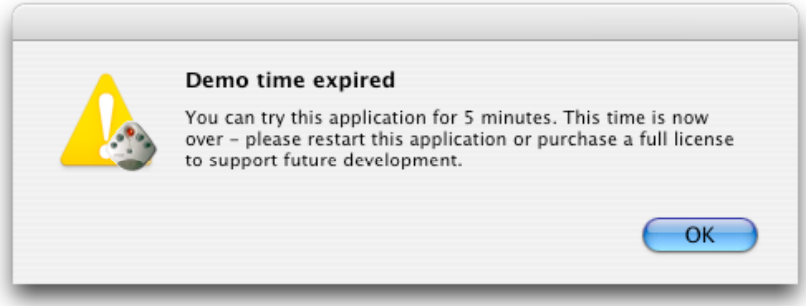
Authorization and Demo Mode

When starting the <Nuke4all> application for the first time you will be asked for a serial number to authorize the program for unrestricted use. In order to do this you must provide a serial number that you can purchase online from the retailers listed on our web site. If you do not have a serial number yet or just want to test drive the program simply select the "Demo" button from the registration screen. You can then try the program for five minutes, which should give you ample time to evaluate all its features and make sure that it works on your system.



The <Nuke4all> registration screen

When the demo time is up, the program will quit. Of course, you can re-start the application for another five minutes.

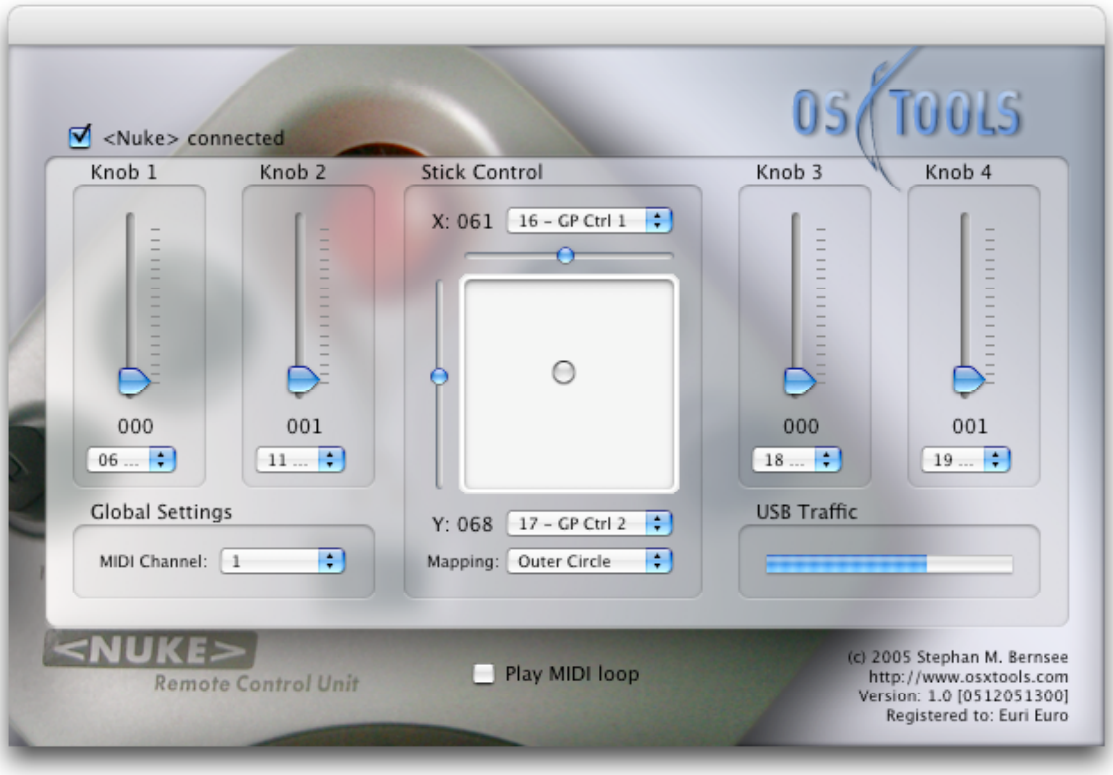


When you're done evaluating the software we recommend you go to the online store and purchase a copy if you like it. We have reduced copy protection to a minimum in order to give you the best performance and working experience with the program. Please do not share your copy of the software. Sharing software will disable us from developing further products and from providing adequate support and updates.

Once you've obtained your serial number, type in your name and serial number exactly as specified on your purchase receipt. Please note that serial numbers consist of numbers 0-9 and characters A-F, no whitespace.

The <Nuke4all> User Interface

The <Nuke4all> user interface is split up in four sliders and one two-dimensional X/Y control. The four sliders are mapped to the rotary knobs on the <Nuke> and the X/Y control is mapped to the orange stick on the <Nuke>.



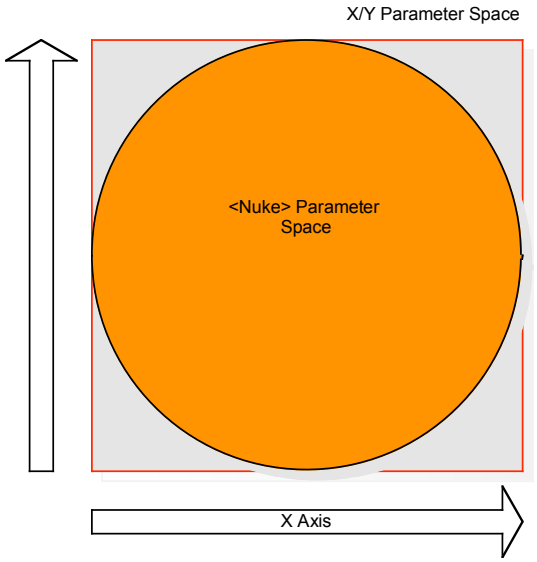
The <Nuke4all> user interface

The sliders are for visualization purposes only and do not send any control change messages when you move them directly with the mouse. They are controlled only by turning the corresponding knob on the <Nuke> remote. Below each slider you will find a popup menu that selects the control change message assigned to this slider. Note that these assignments are remembered across sessions.

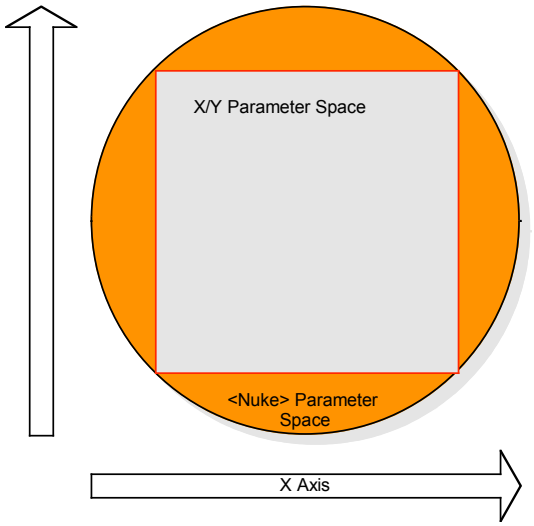
X/Y Control Mapping

The X/Y control also provides a popup called "mapping". Since the control of the stick on the <Nuke> remote is round, there are some "corners" in the parameter space that you cannot reach (the grey areas in the left picture below). The area on the <Nuke> is smaller than the rectangle covering the parameter range for X and Y. To avoid this problem, you can optionally scale the range by selecting the "outer circle" option, which places the X/Y rectangle inside a larger circle.

"Inner Circle" option



"Outer Circle" option



Parameter resolution

The <Nuke> remote hardware provides 8 bits of data resolution. Due to the MIDI control change commands using only 7 bits this resolution is converted down to the MIDI resolution.

Play MIDI Loop checkbox

The "Play MIDI Loop" checkbox has been included as an easy way to test your setup without having a <Nuke> attached to your computer. If enabled, it will send notes to the output port of the application that listens to the <Nuke4all> MIDI port.



If you have a <Nuke> remote control attached to your computer you can also press the button on the interface to play a MIDI chord through your setup.

Connecting and disconnecting the <Nuke>

Since the <Nuke> is a USB device it is perfectly safe to plug and unplug it while your computer is running. <Nuke4all> will automatically detect this and issue a warning to this effect.



The <Nuke> remote was unplugged

Make sure you plug in the <Nuke> again to continue using it.

Note that the MIDI source is not affected by unplugging the <Nuke> so you don't have to change your setup after unplugging the <Nuke>.

! Note: some hosts don't like it when the <Nuke> becomes disconnected and the associated reset message is being sent over MIDI during playback and they might reset their channels settings. Unless prompted by the program we recommend you unplug the <Nuke> only when not using it or when the program <Nuke4all> is not currently running.

Known issues with the <Nuke>

There are several minor issues that exist due to the current software running on the microcontroller on the <Nuke> hardware. These issues cannot be solved in the <Nuke4all> software and need to be worked around.

1. On some systems, the <Nuke> is sometimes not recognized or accessible when <Nuke4all> is started, even though the <Nuke> is plugged in. In that case you might get the following error message:



   Alternatively, the software user interface might simply be greyed out. If this happens, make sure no other software (like the NeuronVS) is currently using the <Nuke> remote control. Click "Ok" if this message is displayed, unplug and re-plug the <Nuke> while the <Nuke4all> application is running and this should solve the problem.

2. Sporadic X/Y messages. The stick control on the <Nuke> is so sensitive that it sometimes reacts to vibrations (and even sound waves) coming in from the environment. If the vibration is strong enough, the <Nuke> will transmit a stick movement, which shows up as an erratic message via MIDI. There is currently no way around this because it is a hardware issue.

3. No stick movements are possible when the <Nuke> button is depressed. This is due to the way button messages are encoded by the <Nuke> remote and there is no way to work around this.

Setup in Cubase SX, Logic and other hosts

Most hosts automatically detect the MIDI source provided by the <Nuke4all> application and provide a way to select it as MIDI source for any of your tracks.
Logic will automatically use the MIDI input from the <Nuke4all> application so you do not have to set up the <Nuke> MIDI Source manually.
In Cubase, the <Nuke> MIDI Source will be selectable from the "In" menu of your track as shown below.



MIDI in selection in Cubase

Note that Logic by default sends the incoming MIDI messages to all output ports. Since the <Nuke4all> also provides a MIDI input port for future enhancements, the MIDI output of the <Nuke> will loop back on itself. This will cause Logic to shut off the MIDI port temporarily due to a buffer overflow. To avoid this from happening, please set the "Port" popup menu for your track to "off".



Set port menu to "off" in Logic

Troubleshooting

Below you can find a small section with frequently asked questions that might help identifying and solving a particular problem.

I am using Logic and when i start Logic, i see tons of MIDI I/O displayed in the Transport bar, going up and down. At some point this stops, but the data from my <Nuke> is also not recognized any longer.

Probably your default Logic song loads with a MIDI track selected in the arrange-window and it's output port is set to "all". This causes a MIDI feedback to the <Nuke>, which blows up the communication. Set the port of the channel to a different value, then open the LogicPro > Preferences > Global dialog and close it again (this will reset MIDI communication)… your <Nuke>should now work as a MIDI controller.

My <Nuke> is not recognized by your app… what am I doing wrong?

The software running on the <Nuke> is...well...let's just say it does some weird things. There's nothing we can do about that (and we didn't write it in the first place), but you can work around this by unplugging the <Nuke> and re-plugging it while the app is running and you'll be fine.

When I rotate one of the rotary encoders (knobs), the <Nuke> sends the MIDI-cc# assigned to it plus the cc# assigned to the stick...what's wrong?

The stick controller is highly sensitive, and interprets the vibrations that turning an encoder induces into the chassis as a stick-movement, and thus sends it's values. This is a hardware design flaw that we can do nothing about.

I can't seem to get the cc# sent by the stick to reach certain values, why is that?

There's a setting called "Mapping" in the Stick Control section of the user interface, you seem to have it set to "Inner Circle". Set it to "Outer Circle" and you'll be ok. There's a "Mouse-Over" info panel that explains what this setting does.

Why can't i change the chord the button on the <Nuke> sends?

Due to a(nother) design flaw in the <Nuke>hardware, it cannot send any control data while this button is pressed, so sending note events is pretty

pointless anyway (except for quick testing whether the <Nuke> does indeed transmit MIDI). In a future update this button will be put to a new, more practical use, so we figured out it's not worth the effort to make the chord editable.

Get Support

If you have any questions on this product or need help using it please contact us at http://www.osxtools.com. Make sure you have your serial number ready when emailing us as we cannot answer your inquiry without a valid serial number.

We are working hard to answer all requests in a timely manner and to solve any open problems as quickly as possible.

Special Thanks

The author would like to thank the following people for their support and contributions to this software (in no particular order): Denis Gökdag of SurroundSFX.com for testing and assistance, Melisande Bernsee for her patience, Sanjay Patel of the Apple Architecture and Performance Group for helping fix bugs in the PerfTools, Bernhard Bouché for providing a <Nuke>, the Apple IORegistryExplorer and USB Prober developers, and Sascha Kujawa at Emagic/Apple for providing me with the opportunity to test and resolve Logic compatibility issues.

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